HAORAN GUO

ghr@mail.ustc.edu.cn | leoguohr.github.io

EDUCATION

University of Science and Technology of China (USTC)

School of Information Science and Technology | Undergraduate | Sep 2024 - Expected Jun 2028

GPA: 3.52/4.3

Selected coursework: Mathematical Analysis, Linear Algebra, Modern Control Theory.

RESEARCH & ENGINEERING PROJECTS

NEXUS: 3D Topology VAE and Vertex Diffusion Jul 15 - Aug 15, 2026

- Implemented topology VAE and point-cloud-conditioned octree vertex diffusion components in PyTorch; built data processing, training recovery, structural evaluation, and visualization workflows.
- Conducted controlled experiments on Fourier coordinate features, Graph/LayerNorm ordering, and face negative sampling to investigate reconstruction and optimization behavior.
- Achieved **99.63% actual Face micro-F1 on 100 fixed training meshes**, with faces classified from predicted edge graphs; five fixed posterior-noise evaluations produced similar scores.

Persona-to-Scene: Multi-Agent Scene Generation and Editing Jul 15 - Aug 15, 2026

- Developed a multi-agent pipeline from persona descriptions to scene content, spatial layouts, and structured generation instructions for interior image generation and event-driven editing.
- Used evidence tracking, schema validation, and paired-image evaluation in end-to-end case studies.

SceneFun3D Track 1: Functionality Segmentation

- Participated with team Night's Watch; prepared the environment and dataset, reproduced a 3D perception baseline, and organized training, inference, and evaluation workflows.
- Investigated functional-region understanding and affordance localization in indoor scenes. Test-set results are listed on the official SceneFun3D benchmark.

RoboGame: RGB-D Perception and Robotic Grasping Jun - Oct 2026

- Built RealSense D435i block detection and 3D localization with depth alignment, IMU tilt compensation, and multi-frame averaging. Integrated camera-to-arm transforms, five-servo inverse kinematics, and suction-grasp control; completed hardware integration testing.

Facial Expression Recognition

- Implemented MobileNetV2 classification on FER2013 and compared accuracy and computational cost with ResNet, EfficientNet, and MAE-DFER.

TECHNICAL SKILLS

Programming: Python, C/C++, Linux/Ubuntu, Git.

Machine learning: PyTorch, training and debugging, evaluation, 3D mesh processing.

Robotics and tools: MuJoCo, STM32, MATLAB.

HONORS & ACTIVITIES

USTC Outstanding Student Scholarship; member of the USTC boxing team.